Characterizing the role of Ing4 in hematopoietic stem cells: an experimental and computational approach

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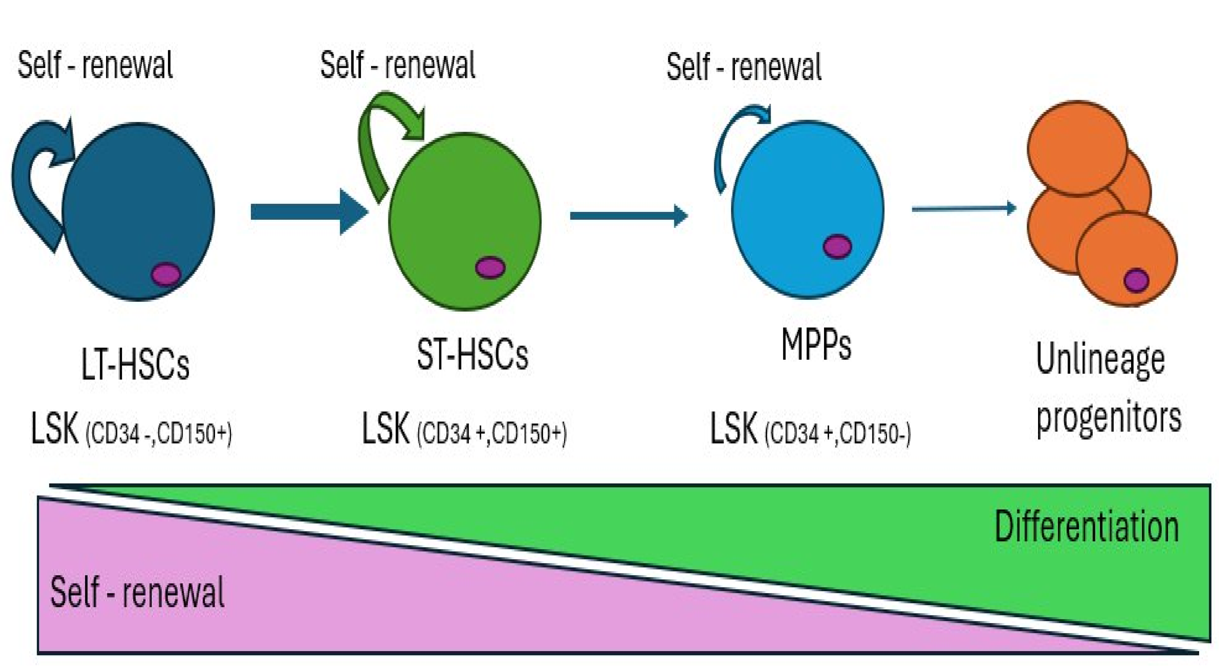
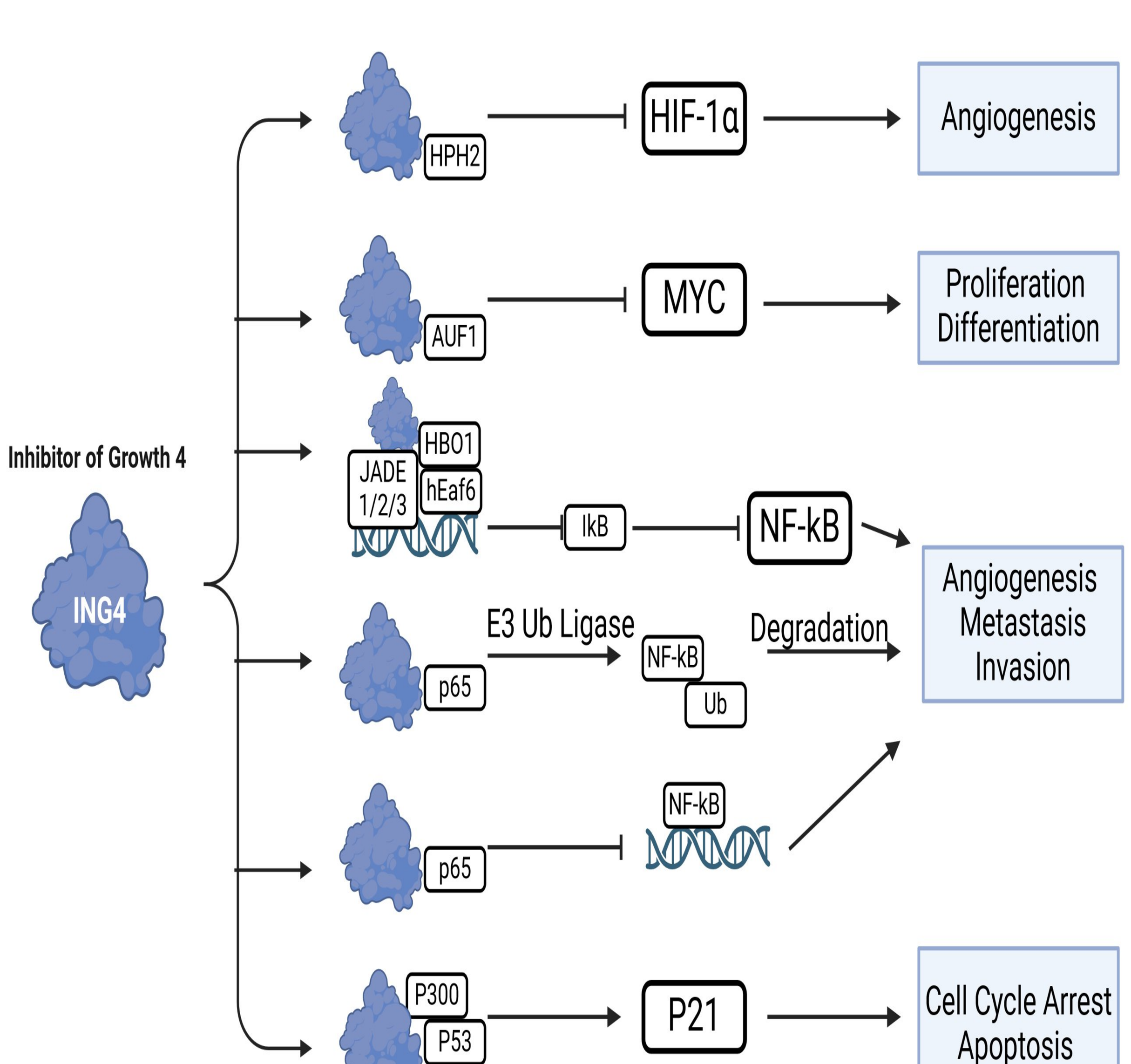
Abstract

Hematopoiesis is a finely regulated process that fluctuates to meet demand, generating fully matured cells from hematopoietic stem cells (HSCs) through controlled self-renewal and differentiation. The inhibitor of growth 4 (Ing4) is a tumor suppressor that has been well characterized in several cancers, but its role in hematopoiesis remains unclear. Our research has shown a pivotal role of Ing4 in HSC regulation. Using an Ing4 knockdown mouse model (Ing4 $^{-/-}$), we observed that HSCs exhibit increased quiescence compared to Wild Type HSCs, yet surprisingly display a transcriptome profile like proliferating cells. Additionally, Ing4 loss enhances the regenerative capacity of HSCs, providing stress resistance against chronic inflammation. These findings suggest Ing4 as a potential candidate for improving hematopoiesis due to robust HSC function, therefore, further investigation into the biological pathways modulating HSC proliferation and differentiation must be performed. To elucidate these pathways, HSCs are extracted from both Ing4 $^{-/-}$ and WT bone marrows and cultured in vitro for seven days to assess how inhibitors targeting different pathways modify HSC proliferation capacity. On the other hand, profound understanding complex systems like hematopoiesis can be challenging, often limited by experimental constraints. To overcome this, we developed an agent-based stochastic computational model to elucidate HSC dynamics in both WT and Ing4 $^{-/-}$ systems. The model integrates experimental data and theoretical work, taking into consideration quiescence, self-renewal, differentiation, and apoptosis. This model will be tested and calibrated until matching the experimental data. After that, it will enable exploration of scenarios that cannot be observed experimentally. Both complementary approaches will aim to enhance our understanding of stem cell biology and contribute to elucidating the characteristics of Ing4-modulated HSCs during bone marrow hematopoiesis.

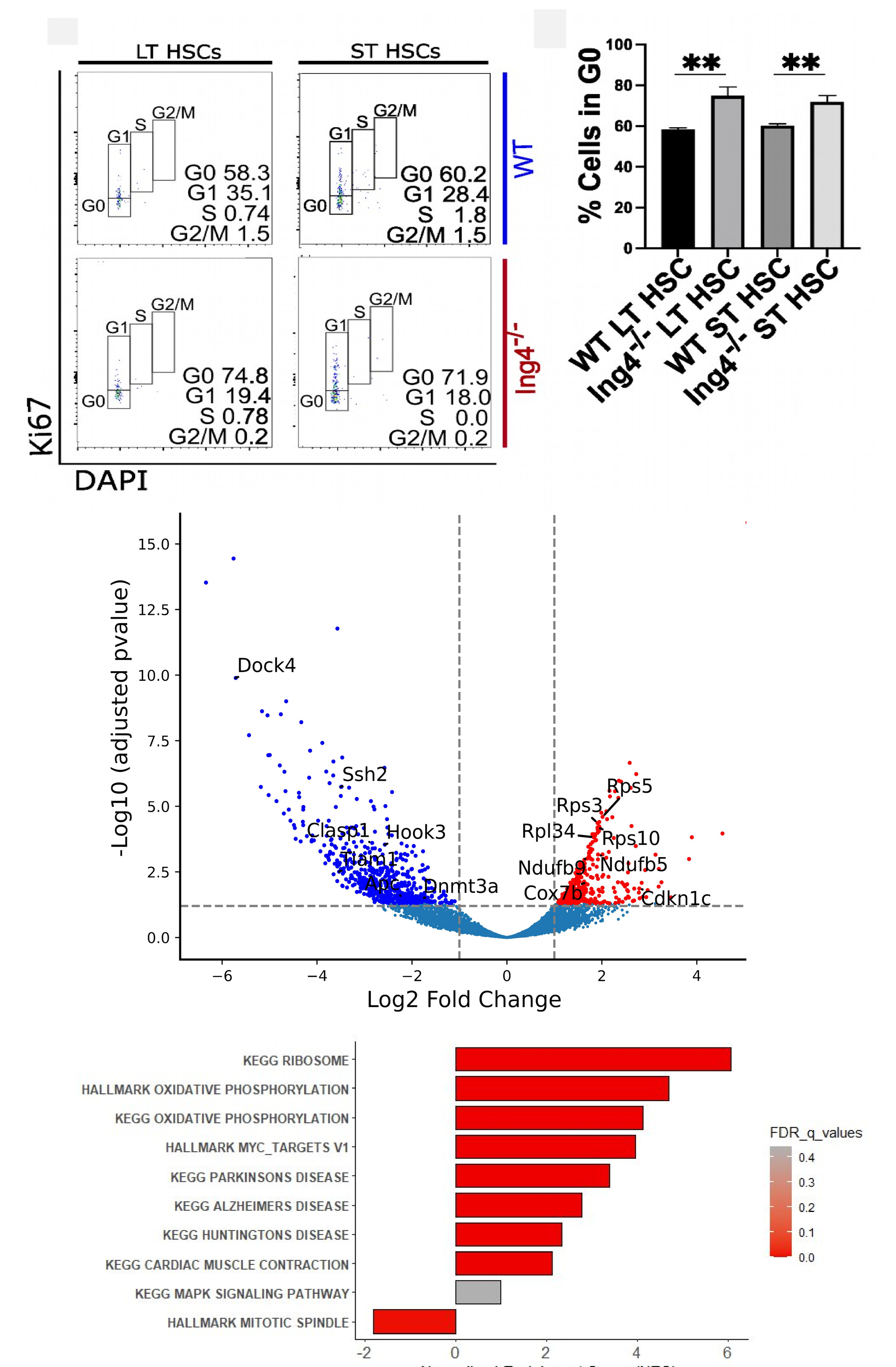
Hematopoietic stem cells are essential for treating hematological diseases



Ing4 is a tumor suppressor that has been characterized in cancer, but it is function in hematopoiesis is not well-known



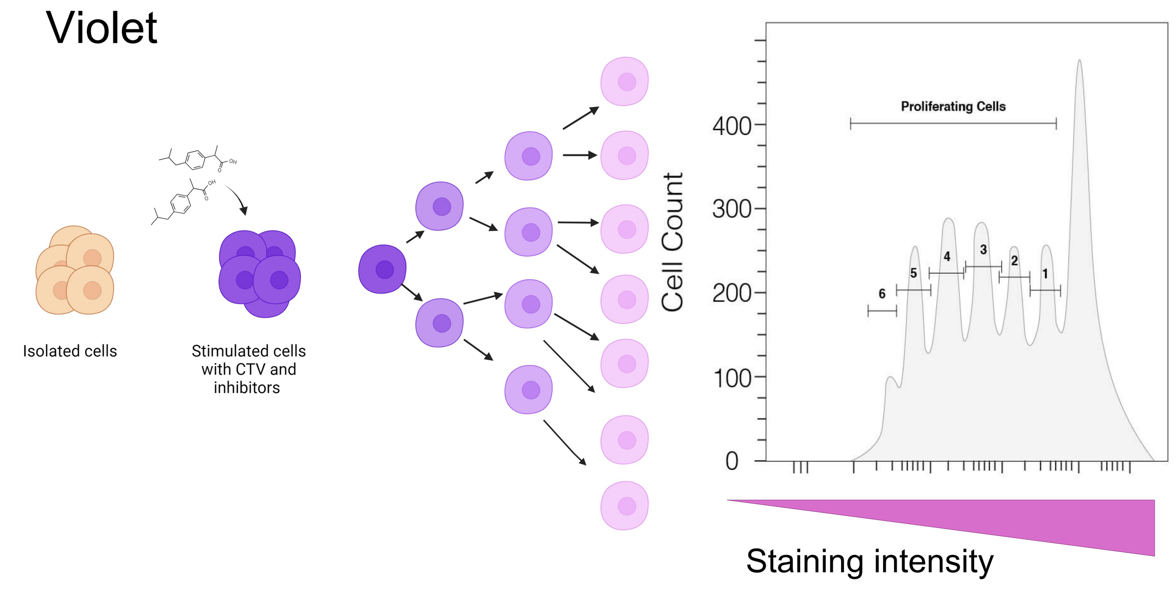
Ing4 $^{-/-}$ HSCs are in quiescent state and poised for activation



Bulk RNA-seq of Ing4 $^{-/-}$ HSCs shows up-regulated and down-regulated genes that are associated with HSCs activation. Surprisingly, Ing4 $^{-/-}$ HSCs are even more quiescence than WT HSCs.

Methodology

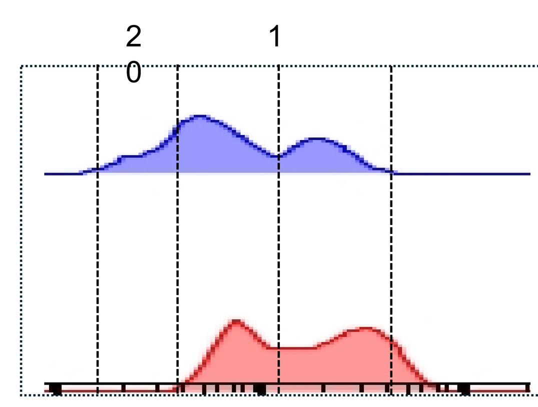
Ing4 $^{-/-}$ HSC ribosomal biogenesis is inhibited by BMH-21 and HSC proliferation is tracked using Cell Trace Violet



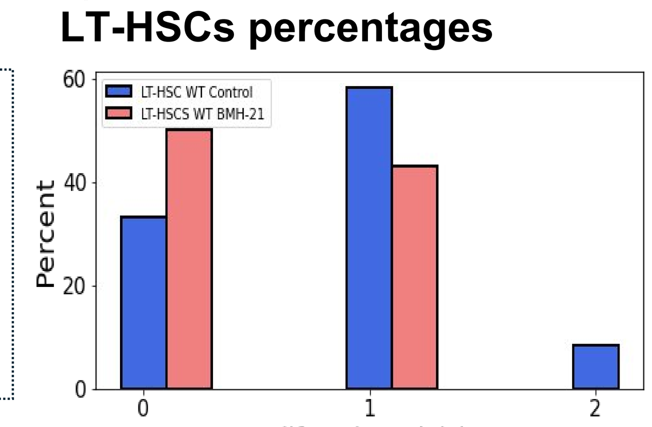
Results

BMH-21 brings back LT-HSCs proliferation to Wild type levels

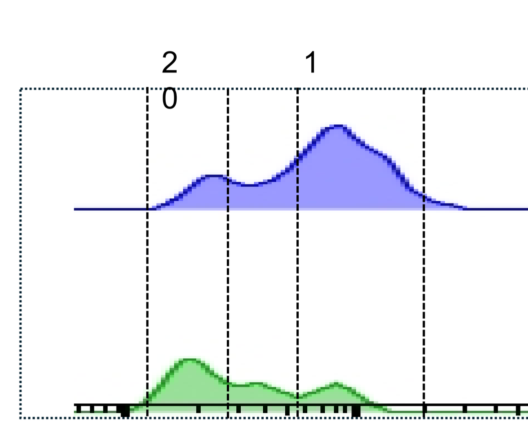
1) WT LT-HSCs divisions



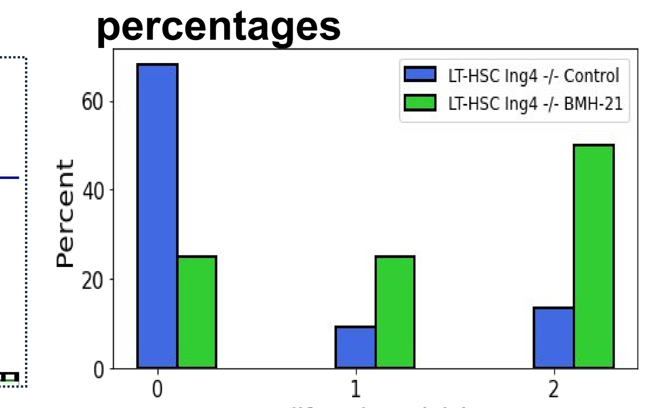
2) WT LT-HSCs vs BMH-21 LT-HSCs percentages



3) Ing4 $^{-/-}$ LT-HSCs divisions

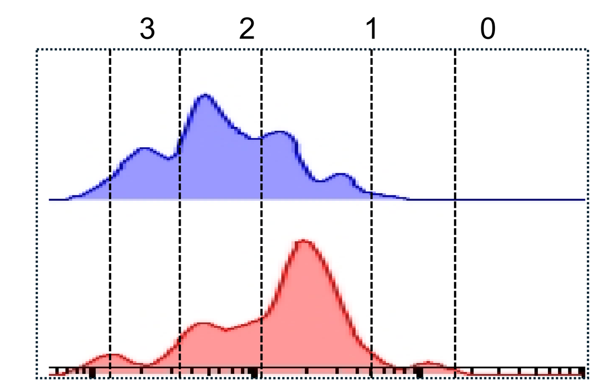


4) Ing4 $^{-/-}$ LT-HSCs vs BMH-21 Ing4 $^{-/-}$ LT-HSCs percentages

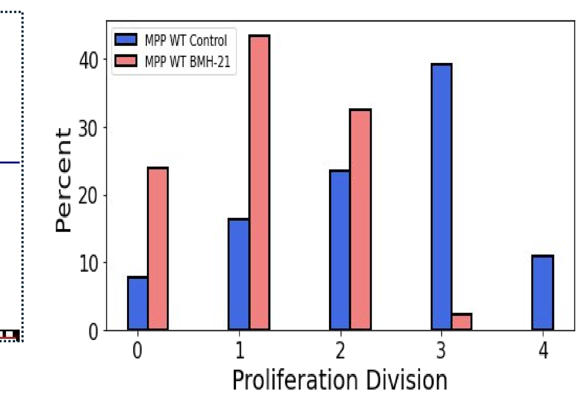


BMH-21 brings back ST-HSCs proliferation to Wild type levels

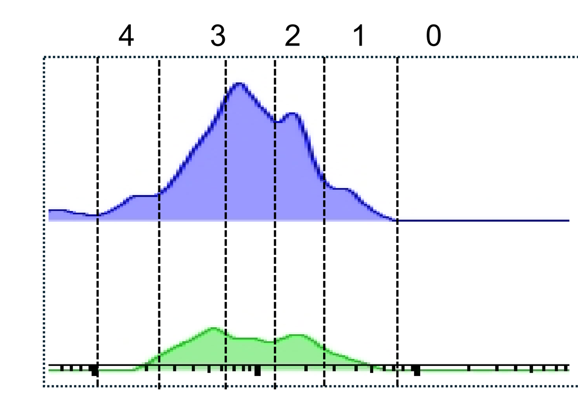
1) WT ST-HSCs divisions



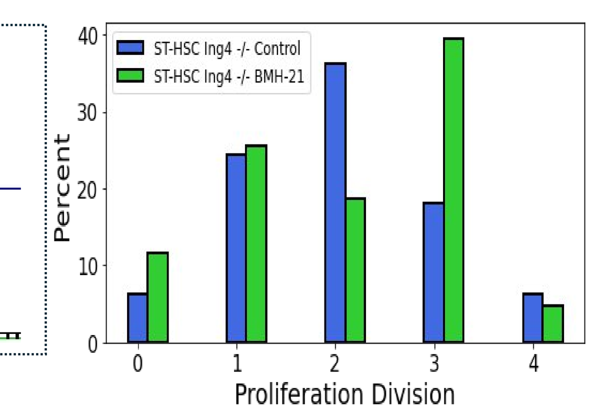
2) WT ST-HSCs vs BMH-21 ST-HSCs percentages



3) Ing4 $^{-/-}$ ST-HSCs divisions

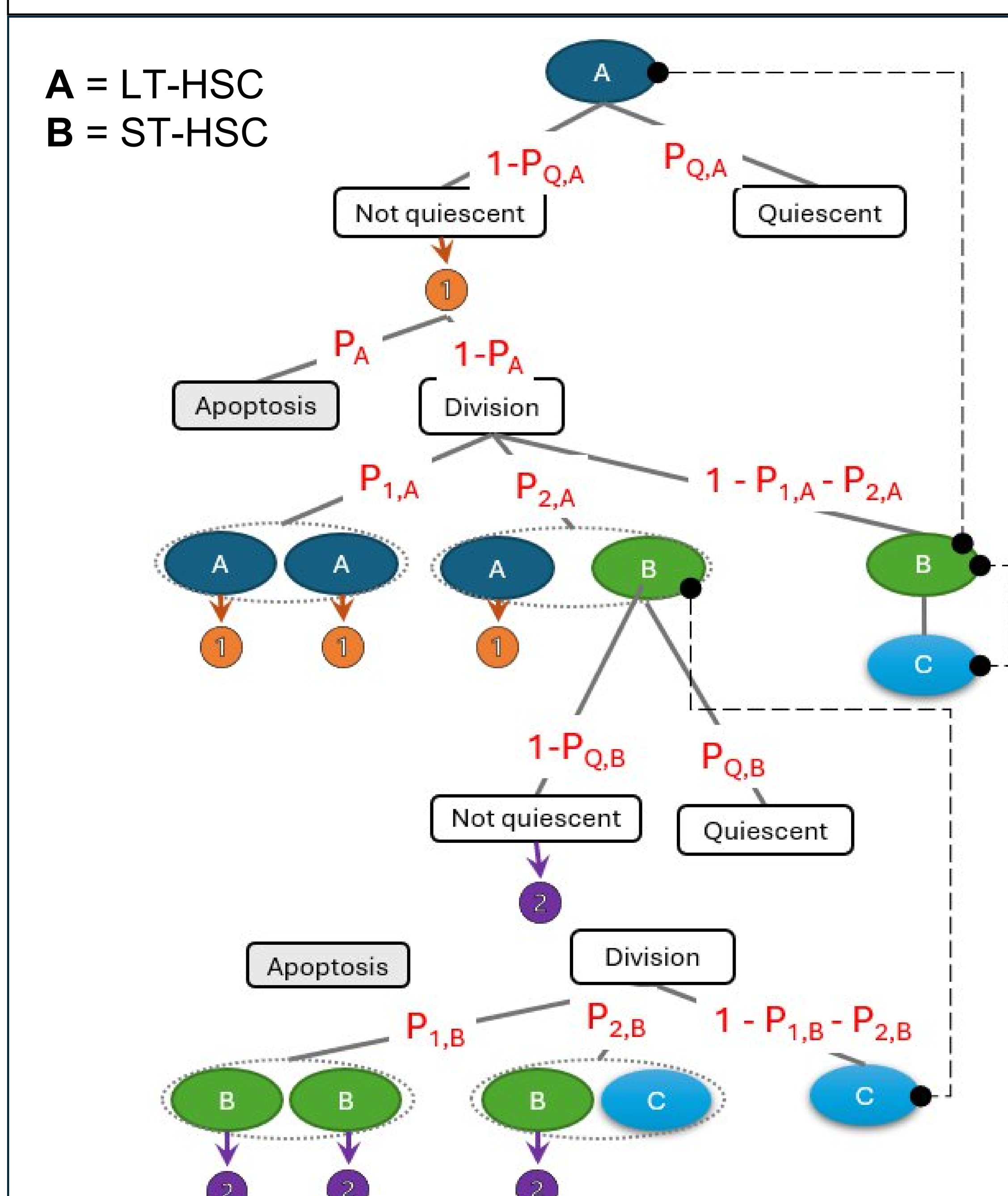


4) Ing4 $^{-/-}$ ST-HSCs vs BMH-21 ST-HSCs percentages



Agent-based stochastic model aiming to describe HSCs dynamics

A = LT-HSC
B = ST-HSC



Conclusions and Future directions

- Ing4 is playing a crucial role in hematopoiesis by influencing HSCs dynamics.
- Ing4 $^{-/-}$ HSC are in quiescent state and expresses genes related to HSC activation.
- Ribosomal biogenesis is a pathway that is upregulated in Ing4 $^{-/-}$ HSCs.
- Inhibition of ribosomal biogenesis in Ing4 $^{-/-}$ LT HSCs and ST-HSCs causes them to exit their quiescent state and proliferate at WT levels.
- We will further investigate another mis-regulate pathways in Ing4 $^{-/-}$ HSCs and the effect of HSCs proliferation as well as differentiation.
- We will test the agent-based model aiming to match the experimental data, and we will explore different biological scenarios related to HSCs dynamics.

Acknowledgments

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